Why axioms?

In the early 20th century, sets were described as "well-defined collections of objects". This leads to contradictions like the Russell's paradox.

Surely the set of all sets that do not belong to themselves is a well-defined collection. Call it Y. So $Y = \{x : x \notin x\}$. Now either $Y \in Y$ or $Y \notin Y$. But each case implies the other (Why?). So we get a contradiction!

Clearly, something has gone wrong. We must be more precise about the notion of sets. This can be done via an axiomatic theory of sets called ZFC (shorthand for Zermelo-Fraenkel set theory with the axiom of choice). As is common in any axiomatic theory (like Euclid's axioms for plane geometry), sets and membership are "primitive notions" and the axioms describe the precise rules to reason with them.

Axioms of ZFC

- **Axiom of empty set**: There is a set with no members.

$$(\exists X)(\forall y)(y \notin X)$$

- **Axiom of extensionality**: Two sets are equal iff they have the same members.

$$(\forall X)(\forall Y)[(X \subseteq Y \& Y \subseteq X) \implies (X = Y)]$$

Extensionality implies that there is a unique empty set which we denote by $\emptyset$ (and later by 0).

Pairing and Union

Most of the ZFC axioms describe how to construct new sets out of old.

- **Axiom of pairing**: For any two sets x, y, there is a set whose members are x, y.

$$(\forall x)(\forall y)(\exists Z)(Z = \{x, y\})$$

- **Axiom of union**: For every family $\mathcal{F}$ of set, there is a set whose members are the members of members of $\mathcal{F}$.

$$(\forall \mathcal{F})(\exists Y)[Y = \{v : (\exists X \in \mathcal{F})(v \in X)\}]$$

We write $\bigcup \mathcal{F}$ to denote the union of the sets in $\mathcal{F}$. If $X_1, X_2, \ldots, X_n$ are sets, we define

$$X_1 \cup X_2 \cup \cdots \cup X_n = \bigcup \{X_1, X_2, \ldots, X_n\}$$

Comprehension scheme

The **axiom of comprehension** says that for any set X and a "first-order property" $\phi(v)$, there is a subset Y of X whose members are precisely those members v of X which satisfy the property $\phi(v)$.

$$(\forall X)(\exists Y)(Y = \{v \in X : \phi(v)\})$$

So axiom of comprehension is a really an axiom scheme as we get one axiom for each "property" $\phi(v)$. We will give a precise definition of "first-order property" later.

Using comprehension

During the course of these lectures, we'll sometimes introduce new sets via the expression $\{x : \phi(x)\}$. As noted before, for some properties $\phi(x)$ (like $x \notin x$) there is no such set. Therefore, on such occasions, one must check that that the axioms of ZFC guarantee the existence of such sets. For example, define the **difference of two sets** by

$$A \setminus B = \{x : x \in A \ \& \ x \notin B\}$$

This is a set since it equals $\{x \in A : x \notin B\}$ which exists by comprehension.

Intersection

If $\mathcal{F}$ is a **nonempty** collection of sets, then we define

$$\bigcap \mathcal{F} = \{y : (\forall X \in \mathcal{F})(y \in X)\}$$

To see that $\bigcap \mathcal{F}$ exists, using the fact that $\mathcal{F} \neq \emptyset$, fix an arbitrary $Z \in \mathcal{F}$ and apply comprehension to conclude that $\bigcap \mathcal{F} = \{v \in Z : (\forall X \in \mathcal{F})(v \in X)\}$ exists. Define

$$X_1 \cap X_2 \cap \cdots \cap X_n = \bigcap \{X_1, X_2, \ldots, X_n\}$$

Two sets are **disjoint** iff their intersection is the empty set. We say that $\mathcal{F}$ is a **disjoint family** iff for every $A \neq B$ in $\mathcal{F}$, $A \cap B = \emptyset$.

Replacement scheme

Suppose X is a set and $\phi(x, y)$ is a "first-order property" such that for every $x \in X$, there is a **unique** set y for which $\phi(x, y)$ holds. Then we can form the set

$$\{y : (\exists x \in X)(\phi(x, y))\}$$

We'll say more on this later when we discuss transfinite recursion.

Power set

The **power set axiom** says that for every set X, there is a set that contains all subsets of X.

$$(\forall X)(\exists Y)(Y = \{S : S \subseteq X\})$$

We denote that power set of X by $\mathcal{P}(X)$.

Natural numbers and the axiom of infinity

Definition (Natural numbers)

- $0 = \emptyset$
- $1 = \{0\}$, $2 = \{0, 1\}$, $3 = \{0, 1, 2\}$ ...
- $n + 1 = n \cup \{n\}$

A set X is **inductive** iff $0 \in X$ and for every $x \in X$, $x \cup \{x\} \in X$. The **axiom of infinity** says that there is an inductive set. We define ω to be the intersection of all inductive sets.

Definition (The set of natural numbers)

$$\omega = \{0, 1, 2, \ldots, n, n + 1, \ldots\}$$

Other axioms

The remaining two axioms are

- **Axiom of choice**
- **Axiom of foundation**

We'll introduce the **axiom of choice** later. For the purposes of this course, we can safely ignore the **axiom of foundation**.

The axioms of ZFC

- **Axiom of empty set**
- **Axiom of extensionality**
- **Axiom of pairing**
- **Axiom of union**
- **Axiom scheme of comprehension**
- **Axiom scheme of replacement**
- **Axiom of power set**
- **Axiom of infinity**
- **Axiom of choice**
- **Axiom of foundation**

Ordered pairs

Definition

The **ordered pair with first coordinate** x **and second coordinate** y *is defined by*

$$(x, y) = \{\{x\}, \{x, y\}\}$$

Note that (x, y) exists by the axiom of pairing. The key property of ordered pairs is the following.

Proposition

If $(x, y) = (a, b)$*, then* $x = a$ *and* $y = b$*.*

The proof is left as an exercise.

Cartesian products

Definition

The **cartesian product** $X \times Y$ *is defined to be the set of all ordered pairs whose first coordinate is in* X *and second coordinate is in* Y*.*

$$X \times Y = \{(x, y) : x \in X \ \& \ y \in Y\}$$

Note that $X \times Y$ is a subset of $\mathcal{P}(\mathcal{P}(X \cup Y))$ which exists by the pairing, union and power set axioms. So the existence of

$$X \times Y = \{v \in \mathcal{P}(\mathcal{P}(X \cup Y)) : (\exists x \in X)(\exists y \in Y)(v = (x, y))\}$$

follows from comprehension.

Relations

A **relation** R is a set of ordered pairs. If R is a relation, then

- $\text{dom}(R) = \{x : (\exists y)((x, y) \in R)\}$
- $\text{range}(R) = \{y : (\exists x)((x, y) \in R)\}$

We say that R is a **relation from** A **to** B iff $R \subseteq A \times B$. Note that every relation R is a relation from $\text{dom}(R)$ to $\text{range}(R)$. We say that R is a **relation on** A iff R is a relation from A to A.
Notation: If R is a relation, we sometimes write xRy (read x is R-related to y) instead of $(x, y) \in R$.

Functions

F is a **function** iff F is a relation and for every $x \in \text{dom}(F)$, there is a unique $y \in \text{range}(F)$ such that $(x, y) \in F$. We write $F(x) = y$ instead of $(x, y) \in F$. We say that F **is a function from** A **to** B, and write $F : A \to B$, iff F is a function, $\text{dom}(F) = A$ and $\text{range}(F) \subseteq B$.
Suppose $F : A \to B$. We say that

- F is **injective** (one-one) if for every $x \neq y$ in A, $F(x) \neq F(y)$.
- F is **surjective** (onto) if $\text{range}(F) = B$
- F is **bijective** iff it is both injective and surjective.

If $f : A \to B$ and $g : B \to C$, then $g \circ f : A \to C$ defined by $(g \circ f)(x) = g(f(x))$ is the **composition** of f with g.
If $f : A \to B$ is a bijection, the **inverse** of f is the function $f^{-1} : B \to A$ defined by

$$f^{-1}(y) = x \iff f(x) = y$$

Restrictions, Images/preimages, Finite/Infinite

Suppose $F : A \to B$, $X \subseteq A$ and $Y \subseteq B$.

- The **restriction of F to X**, denoted $F \restriction X$ is defined by: $\text{dom}(F \restriction X) = X$ and for every $x \in X$, $(F \restriction X)(x) = F(x)$.
- The **image of X** under F is $F[X] = \{F(x) : x \in X\}$.
- The **preimage of Y with respect to F** is

$$F^{-1}[Y] = \{x \in A : F(x) \in Y\}$$

A set X is **finite** iff for some natural number n, there exists a bijection $f : n \to X$. Otherwise, it is **infinite**.

Isomorphism

Definition (Isomorphism)

Suppose R, S are relations and A, B are sets. We say that (A, R) **is isomorphic to** *(B, S) and write $(A, R) \cong (B, S)$ iff there is a bijection $f : A \to B$ such that for every $x, y \in A$, xRy iff $f(x)Sf(y)$.*

Equivalence relations and partitions

We say that R is an **equivalence relation** on A iff R is a relation from A to A which satisfies the following.

- **Reflexive** For every $a \in A$, aRa.
- **Symmetric** If aRb, then bRa.
- **Transitive** If aRb and bRc, then aRc.

We say that $\mathcal{F}$ is a **partition** of A iff $\mathcal{F}$ is a disjoint family and $\bigcup \mathcal{F} = A$.

Exercise

Suppose R is an equivalence relation on A. For each $a \in A$, define the R-equivalence class of a by $[a] = \{b \in A : aRb\}$. Then $\{[a] : a \in A\}$ is a partition of A.

Linear orderings

Definition

A **linear ordering** *is a pair* $(A, \prec)$ *such that* A *is a nonempty set and* $\prec$ *is a binary relation on* A *that satisfies the following.*

- **Irreflexive:** *For every* $a \in A$, $\neg(a \prec a)$ *(*$\neg$ *denotes negation).*
- **Transitive:** *If* $a \prec b$ *and* $b \prec c$*, then* $a \prec c$*.*
- **Total:** *For every* $a, b \in A$ *if* $a \neq b$*, then either* $a \prec b$ *or* $b \prec a$*.*

If $(A, \prec)$ is a linear ordering, we define the relation $\preceq$ on A by

$$a \preceq b \iff (a \prec b \text{ or } a = b)$$

If $(A, \prec)$ is a linear ordering and $x \in A$, we define the set if **predecessors** of x in $(A, \prec)$ by $\text{pred}(A, \prec, x) = \{y \in A : y \prec x\}$.